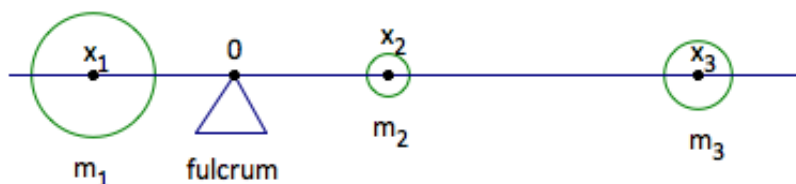


8.3 Moments and Centers of Mass

Masses Along a Line

Many structures and mechanical systems behave as if their masses were concentrated at a single point, called the _____.

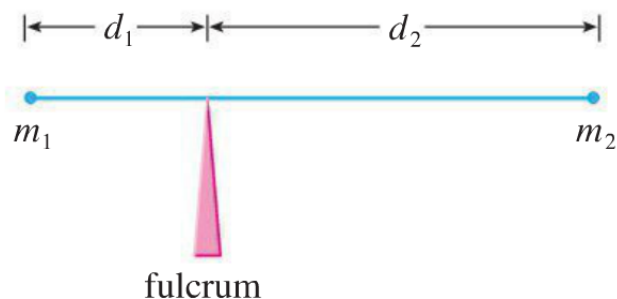


Consider the picture above. The x -axis is supported by a _____ at the origin. Is there any way that the system can balance?

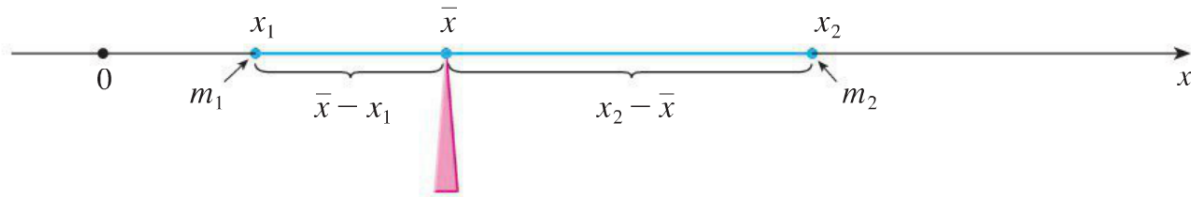
What factors would this system's balance depend on?

- 1.
- 2.

When will the following system be balanced?



What if we wish to determine this in relation to the x -coordinates of the objects in our system?



Definition

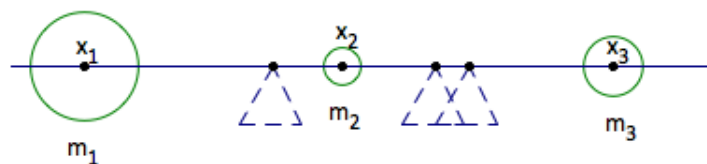
We say that the _____ of a system about the origin is given by

$$\sum_{k=1}^n m_k x_k,$$

for a system with n objects, where m_i represents each objects mass and x_i represents the signed distance from the origin.

We say that each $m_i x_i$ is a _____.

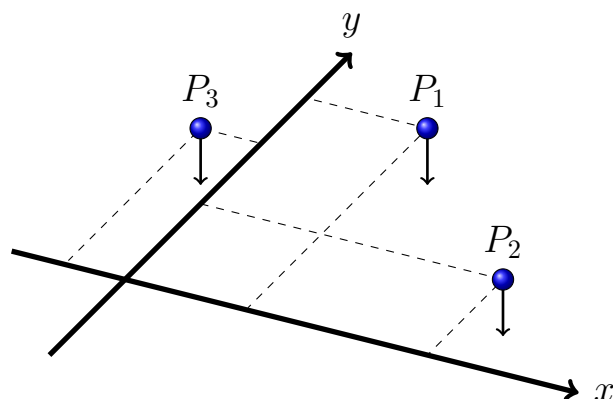
Given a system, we want to know where to place the _____ to make the system balance.



We call this location _____, the system's center of mass.

Masses Along a Plane

We often consider the moments about two axes of objects in two (or three) dimensions.



How far is $P_1 = (x_1, y_1)$
from the x -axis?

How far is $P_1 = (x_1, y_1)$
from the y -axis?

What is the total moment of the whole system about the x -axis?

$$M_x =$$

What is the total moment about the y -axis?

$$M_y =$$

How would we find the center of mass?

$$\bar{x} =$$

$$\bar{y} =$$

We can then draw in the lines of $x = \bar{x}$ and $y = \bar{y}$ and the system is balanced about these lines.

We call $(\bar{x}, \bar{y})$ the _____ for the system.

Example 1

Consider a point-mass system that consists of points of masses $m_1 = 4$, $m_2 = 8$, $m_3 = 3$, $m_4 = 2$ located at:

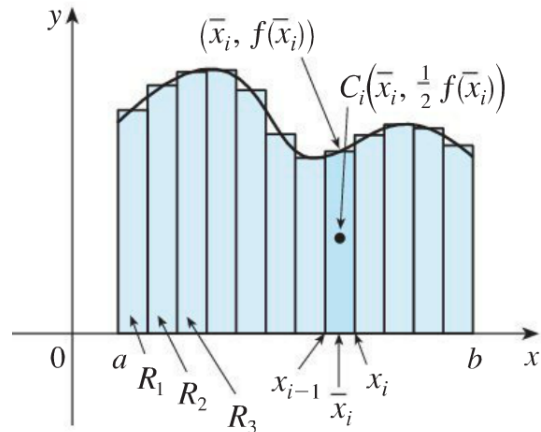
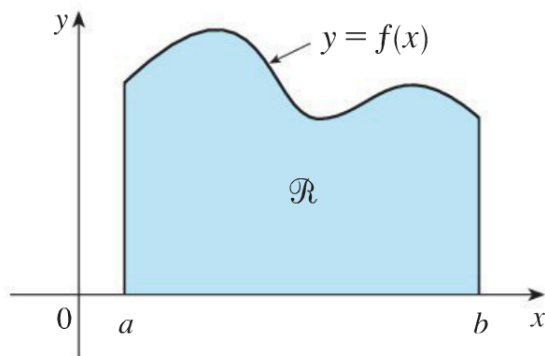
$$P_1 = (-2, 3) \quad P_2 = (2, -6) \quad P_3 = (7, -3) \quad P_4 = (5, 1)$$

Calculate moments M_x , M_y and locate the center of mass $(\bar{x}, \bar{y})$.

Thin, Flat Plates

Sometimes it is necessary to find the center of mass of a thin, flat plate called a _____ with uniform density, ρ that occupies a region $\mathcal{R}$ of the plane.

The center of mass of a region with uniform density is called the _____ of $\mathcal{R}$.



How do we calculate mass?

Let's use vertical strips to approximate the moments and center of mass

center of mass

length

width

area

mass

distance from
c.m. to axis

Formulae

Moment about the x -axis $M_x =$

Moment about the y -axis $M_y =$

Mass $M =$

Center of mass $\bar{x} =$

$\bar{y} =$

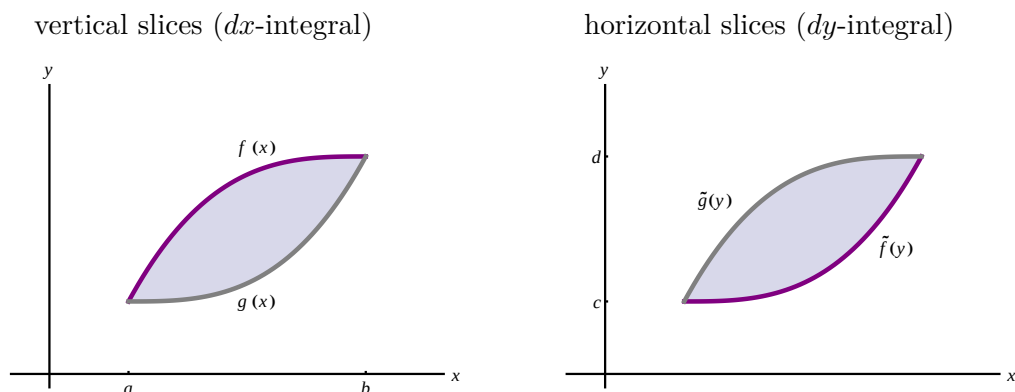
Example 2

Find the centroid of the region bounded above by the unit circle and below by the x -axis.

NOTE: The remaining material in these lecture notes is supplemental material. While it is suggested to continue with the lecture to further your understanding, the following material in this section is not required.

Solids Bounded by Curves (not required)

To calculate the coordinates of center of mass we slice the region (or lamina) into strips:



Location of the center of mass of each strip is

$$\bar{x}_i =$$

$$\bar{x}_i =$$

$$\bar{y}_i =$$

$$\bar{y}_i =$$

Mass of the strip:

$$m_i =$$

$$m_i =$$

Moment that each strip creates about the x -axis

$$M_{x,i} =$$

$$M_{x,i} =$$

Moment that each strip creates about the y -axis

$$M_{y,i} =$$

$$M_{y,i} =$$

By taking limits we obtain integral formulae for calculating the moments

$$M_x =$$

$$M_x =$$

$$M_y =$$

$$M_y =$$

and total mass:

$$M =$$

$$M =$$

The final formulae for coordinates of center of mass are given below.

Centers of Mass

$$\bar{x} = \frac{M_y}{M} = \frac{\rho \int_a^b x(f(x) - g(x)) dx}{\rho \int_a^b f(x) - g(x) dx} = \frac{\frac{\rho}{2} \int_c^d \tilde{f}^2(y) - \tilde{g}^2(y) dy}{\rho \int_c^d \tilde{f}(y) - \tilde{g}(y) dy}$$

$$\bar{y} = \frac{M_x}{M} = \frac{\frac{\rho}{2} \int_a^b f^2(x) - g^2(x) dx}{\rho \int_a^b f(x) - g(x) dx} = \frac{\rho \int_c^d y(\tilde{f}(y) - \tilde{g}(y)) dy}{\rho \int_c^d \tilde{f}(y) - \tilde{g}(y) dy}$$

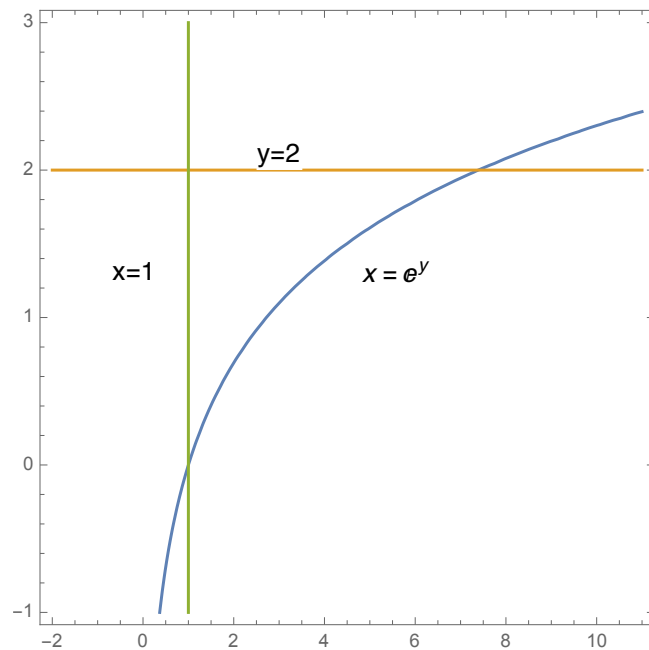
Can we simplify these formulae?

Example 3 (extra)

Let R be the lamina with density $\rho = \frac{1}{3}$ in the region in the xy -plane bounded by $y = \frac{4}{x}$, $x = 2$, $y = \frac{1}{4}$. Find the center of mass.

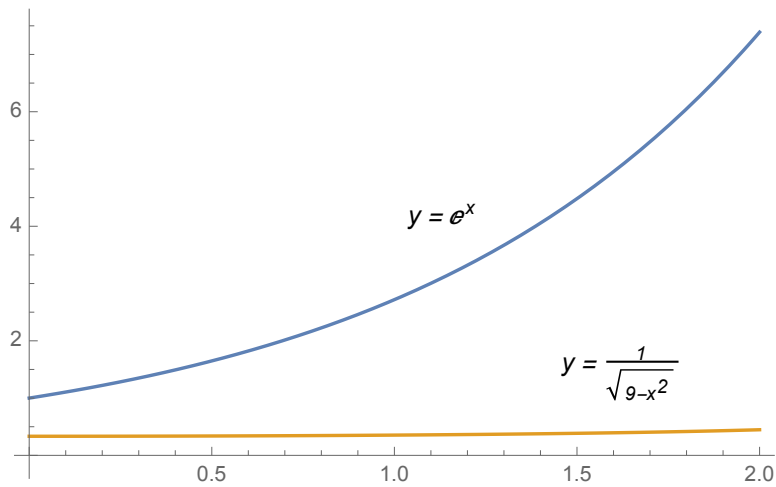
Example 4 (extra)

Let R be the lamina with density $\rho = 3$ in the region in the xy -plane bounded by $x = e^y$, $x = 1$, and $y = 2$. Find the moment M_y of the lamina.



Extra Example

Find the centroid of the region bounded by the curves $y = e^x$, $y = \frac{1}{\sqrt{9-x^2}}$, $x = 2$, and the y -axis.



Example 5 (extra)

Let R be the lamina with density $\rho = \frac{1}{3}$ in the region in the xy -plane bounded by $y = 3x$, $x = 3$, and the x -axis.

(a) Set up integrals for M_x and M_y using slices perpendicular to the y -axis.

(b) Set up integrals for M_x and M_y using slices perpendicular to the x -axis.

(c) Find the center of mass.